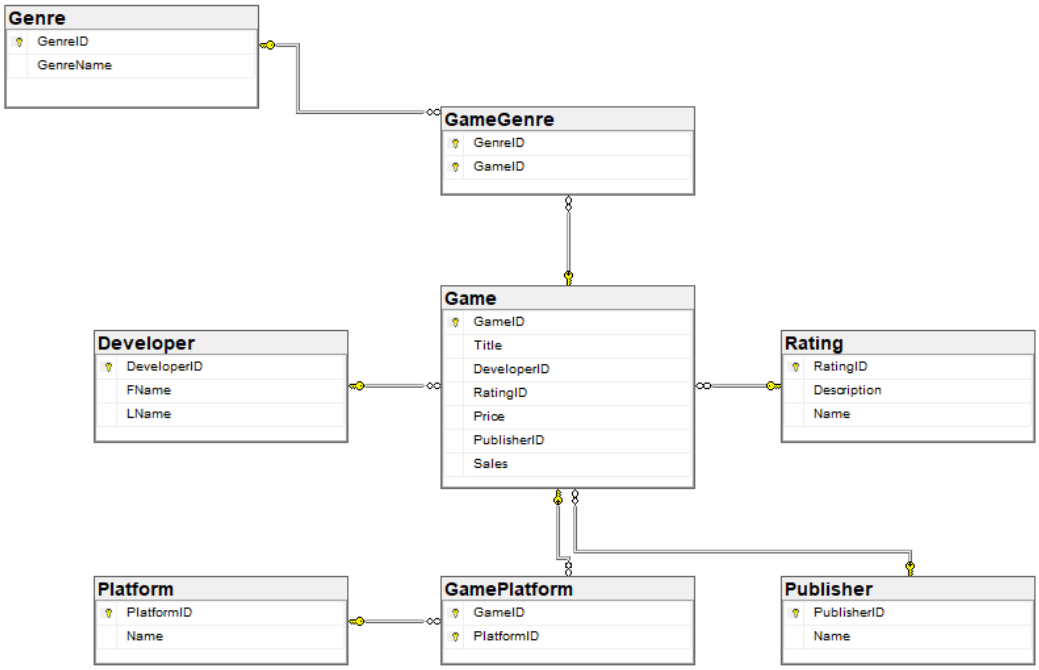
# Videogame Database




When creating the videogame database, we first started generating factors of a videogame. We then broke down each factor and added the necessary descriptions that make the factors work. We put each factor into a table with its corresponding attributes. Each game factor had to have an ID so that we could reference the tables to one another. We had to create two bridge tables to make the many-to-many relationships work such as “GamePlatform” and “GameGenre”. All the other factors were one-to-many or one-to-one relationships, so they didn’t need a bridge table. We laid all our tables together so we could visualize our database image.

After we had our tables set up, we needed data. We created multiple spreadsheets for each table in Excel so we could fill the tables with data. We looked up data using the internet and plugged in the information that we found in the corresponding spots. For example, we looked up different videogame titles and plugged names like “Minecraft” or “Valorant” under Title in our Game table. Once we got all our data in our spreadsheet, we were then able to import the Excel document into SQL Server Management Studio. SQL Server made sure our data on the spreadsheet lined up with our database design and populated all the tables with the data making our database complete.

One of the biggest challenges we had was importing our data from the Excel document. When going to import the data, SQL Server checks to see if the data we are importing matches with the perimeters we have set for our tables. In many cases, we had either the incorrect data type, spelling errors, or having to include an identity insert. We went back and forth with trial-and-error runs until we got it to work. Other than that, everything else went smoothly.

# Videogame Database

## Populated Tables

Game Table:

| | GameID | Title | DeveloperID | RatingID | Price | PublisherID | Sales |
|---|---|---|---|---|---|---|---|
| 1 | 1 | Minecraft | 1 | NULL | 25.00 | 31 | 5950000000.00 |
| 2 | 2 | Terraria | 2 | NULL | 10.00 | 32 | 440000000.00 |
| 3 | 3 | Skyrim: Elder Scrolls V | 2 | NULL | 60.00 | 13 | 3600000000.00 |
| 4 | 4 | Fortnite | 4 | NULL | 0.00 | 30 | 0.00 |
| 5 | 5 | Grand Theft Auto V | 5 | NULL | 60.00 | 17 | 11100000000.00 |
| 6 | 6 | Rocket League | 5 | NULL | 20.00 | 30 | 600000000.00 |
| 7 | 7 | Rust | 7 | NULL | 40.00 | 2 | 480000000.00 |
| 8 | 8 | Need for Speed: Hot Pursuit | 9 | NULL | 60.00 | 3 | 300000000.00 |
| 9 | 9 | Counter-Strike: Global Offensive | 10 | NULL | 0.00 | 2 | 0.00 |
| 10 | 10 | Valorant | 7 | NULL | 0.00 | 21 | 0.00 |
| 11 | 11 | Plants vs. Zombies: Garden Warfare | 6 | NULL | 30.00 | 3 | 2400000000.00 |
| 12 | 12 | Clash of Clans | 5 | NULL | 0.00 | 1 | 0.00 |
| 13 | 13 | Call of Duty: Black Ops 2 | 11 | NULL | 60.00 | 5 | 174000000.00 |
| 14 | 14 | Mario Party 8 | 12 | NULL | 50.00 | 2 | 443000000.00 |
| 15 | 15 | Bloons Tower Defense 6 | 15 | NULL | 10.00 | 2 | 100000000.00 |
| 16 | 16 | Farming Simulator | 14 | NULL | 30.00 | 3 | 120000000.00 |
| 17 | 17 | Madden 24 | 15 | NULL | 60.00 | 3 | 240000000.00 |
| 18 | 20 | NBA 2k 24 | 15 | NULL | 60.00 | 1 | 480000000.00 |
| 19 | 21 | Halo: Reach | 15 | NULL | 60.00 | 22 | 570000000.00 |
| 20 | 22 | Pac-Man | 2 | NULL | 0.25 | 1 | 14000000000.00 |

The Game table gives the corresponding GameID number to the videogame, the title of the game, the DeveloperID number to reference to the Developer table, the RatingID number to reference to the Rating table however null is allowed, the price of the game, the PublisherID number to reference to the Publisher table, and the current sales of each game.

Developer Table:

| | DeveloperID | FName | LName |
|---|---|---|---|
| 1 | 1 | Eric | Barone |
| 2 | 2 | Toru | Iwatani |
| 3 | 3 | Allan | Alcorn |
| 4 | 4 | Masahiro | Sakurai |
| 5 | 5 | Feargus | Urquhart |
| 6 | 6 | Tim | Cain |
| 7 | 7 | Will | Wright |
| 8 | 8 | John | Romero |
| 9 | 9 | Shigeru | Miyamoto |
| 10 | 10 | Hidetaka | Miyazaki |
| 11 | 11 | Andrew | Spinks |
| 12 | 12 | Markus | Persson |
| 13 | 13 | Brandon | Beck |
| 14 | 14 | Todd | Howard |
| 15 | 15 | Jason | West |

The Developer table has a list of different developers separated by first and last names and each has a corresponding DeveloperID number.

# Videogame Database

GameGenre Table:

| | GenreID | GameID |
|---|---|---|
| 1 | 1 | 1 |
| 2 | 1 | 2 |
| 3 | 1 | 3 |
| 4 | 1 | 4 |
| 5 | 1 | 5 |
| 6 | 1 | 12 |
| 7 | 2 | 3 |
| 8 | 2 | 5 |
| 9 | 2 | 17 |
| 10 | 3 | 2 |
| 11 | 3 | 22 |
| 12 | 4 | 3 |
| 13 | 4 | 5 |
| 14 | 5 | 6 |
| 15 | 5 | 17 |
| 16 | 6 | 1 |
| 17 | 6 | 2 |
| 18 | 8 | 2 |

This GameGenre bridge table allows many games to have many genres by using the GameID's number and the GenreID's number.

GamePlatform Table:

| | GameID | PlatformID |
|---|---|---|
| 1 | 1 | 7 |
| 2 | 1 | 8 |
| 3 | 1 | 9 |
| 4 | 1 | 10 |
| 5 | 1 | 11 |
| 6 | 1 | 12 |
| 7 | 1 | 13 |
| 8 | 1 | 14 |
| 9 | 2 | 1 |
| 10 | 2 | 7 |
| 11 | 2 | 8 |
| 12 | 2 | 9 |
| 13 | 2 | 10 |
| 14 | 2 | 11 |
| 15 | 2 | 12 |

Like the GameGenre bridge table, the GamePlatform table allows many games to have many publishers by using the GameID's number and the PlatformID's number.

# Videogame Database

Genre Table:

| | GenreID | GenreName |
|---|---|---|
| 1 | 1 | Action Adventure |
| 2 | 2 | RPG |
| 3 | 3 | Platformer |
| 4 | 4 | Story |
| 5 | 5 | Sports |
| 6 | 6 | Adventure |
| 7 | 7 | Metroidvania |
| 8 | 8 | Fantasy |
| 9 | 9 | Shooter |
| 10 | 10 | Puzzle |
| 11 | 11 | Fighting |
| 12 | 12 | Survival |
| 13 | 13 | Battle Royale |
| 14 | 14 | Rhythm |
| 15 | 15 | Horror |
| 16 | 16 | Interactive |
| 17 | 17 | Sandbox |
| 18 | 18 | Raceing |
| 19 | 19 | Tower Defence |

The Genre table gives a list of many different genres a game could have and matches each genre to a corresponding GenreID number.

Platform Table:

| | PlatformID | Name |
|---|---|---|
| 1 | 1 | NES |
| 2 | 2 | GameCube |
| 3 | 3 | Nintendo 3DS |
| 4 | 4 | Nintendo 2DS |
| 5 | 5 | Playstation1 |
| 6 | 6 | Playstation 2 |
| 7 | 7 | Playstation 3 |
| 8 | 8 | Playstation 4 |
| 9 | 9 | Playstation 5 |
| 10 | 10 | Xbox Original |
| 11 | 11 | Xbox 360 |
| 12 | 12 | Xbox Series S |
| 13 | 13 | Xbox Series X |
| 14 | 14 | Xbox One |
| 15 | 15 | PSP |
| 16 | 16 | Super Nintendo |

The Platform table lists multiple platforms games can be played on and matches each platform to a corresponding PlatformID number.

# Videogame Database

Publisher Table:

| | PublisherID | Name |
|---|---|---|
| 1 | 1 | 2K games |
| 2 | 2 | Nintendo |
| 3 | 3 | Electronic Arts |
| 4 | 4 | Activision |
| 5 | 5 | Activision Blizzard |
| 6 | 6 | Apple |
| 7 | 7 | Armor Games |
| 8 | 8 | Atari inc |
| 9 | 9 | Atari games |
| 10 | 10 | Atlus |
| 11 | 11 | BBC Multimedia |
| 12 | 12 | Ubisoft |
| 13 | 13 | Bethesda Softworks |
| 14 | 14 | Capcom |
| 15 | 15 | Bandai Namco |
| 16 | 16 | Big Fish Studios |

The Publisher Table lists different publishers of games and matches each publisher name to a corresponding PublisherID number.

Rating Table:

| | RatingID | Description | Name |
|---|---|---|---|
| 1 | 1 | All Ages | Everyone |
| 2 | 2 | Eveyone 10+ | Everyone 10+ |
| 3 | 3 | 13+ | Teen |
| 4 | 4 | 17+ | Mature |
| 5 | 5 | 18+ | Adults Only |
| 6 | 6 | Unknown | Rating Pending |

The Rating table has the ratings from the ESRB (Entertainment Software Rating Board) to give the games the correct rating. They each have a description of who is recommended to play the game and have a corresponding RatingID number that can be called to label a certain game.